3. A priori error bounds.

3.1. The marginals. For a fixed dictionary of degree K_2 we assume the moment bounds (T),

(3.1)

$$\mathbb{E}\|\widehat{\boldsymbol{\theta}}^r\|^2 = O(1), \qquad \mathbb{E}\,\mathbb{E}_{\pi_{\boldsymbol{n}}}\Big[(1+\|\mathcal{Z}\|+\|\widehat{\mathcal{Z}}\|)^{2(K_2-1)}\|\widehat{\mathcal{Z}}-\mathcal{Z}\|^2\Big] = O\bigg(\frac{d(\log N)^{K_2}}{N}\bigg).$$

where the outer expectation is over the estimation sample and the inner one over a fresh residual. A mean-value bound for the polynomial $\widehat{\mathcal{L}}_r$ and the Cauchy–Schwarz inequality give $\mathbb{E}|R_N| = O(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2})$ (Appendix A). Combining this with (M) gives

$$\mathbb{E}[\mathcal{E}_{\mathrm{marg}} - \log \widehat{Z}_{\mathrm{comp}}] \;=\; O\big(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2}\big), \tag{3.2}$$

and $\mathcal{E}_{\mathrm{marg}} - \log \widehat{Z}_{\mathrm{comp}} = O_p(\sqrt{d}\,(\log N)^{K_2/2}N^{-1/2})$. The constants depend on d and the dictionary. We compute the remaining correction $\log \widehat{Z}_{\mathrm{comp}}$ for each estimate and do not bound it a priori.

3.2. The subspace. The Stage-1 bounds follow from the next inequality, in which $\mathcal{S}'$ is an arbitrary score field.

Proposition 3.1 (Stage-1 bounds). *Let $\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}$ be the estimated Stage-1 basis of* (3.5) *and $\widehat{\boldsymbol{P}}_r := \widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}\top}$. Under Assumption 2.1 of Part I, for every score field $\mathcal{S}' \in L^2(\pi_{\boldsymbol{Z}};\mathbb{R}^d)$ with second-moment matrix $\boldsymbol{C}' := \mathbb{E}_{\pi_{\boldsymbol{Z}}}[\mathcal{S}'\mathcal{S}'^{\top}]$,*

$$\sqrt{2\,\mathcal{E}_{\mathrm{proj}}} \;\le\; \|\mathcal{S}-\mathcal{S}'\|_{L^2(\pi_{\boldsymbol{Z}})} \;+\; \sqrt{\mathrm{Tr}\big((\boldsymbol{I}-\widehat{\boldsymbol{P}}_r)\boldsymbol{C}'\big)}. \tag{3.3}$$

For $\mathcal{S}' = \mathcal{S}$, the score-error term in (3.3) *vanishes. Squaring the remaining inequality and dividing by two yields the oracle bound*

$$\mathcal{E}_{\mathrm{proj}} \;=\; \mathcal{D}_{\mathrm{KL}}\big(\pi_{\boldsymbol{n}}\,\|\,\pi_{\boldsymbol{n}}(\,\cdot\,;\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}})\big) \;\le\; C_1 \;:=\; \tfrac{1}{2}\,\mathrm{Tr}\big((\boldsymbol{I}-\widehat{\boldsymbol{P}}_r)\boldsymbol{C}\big), \tag{3.4}$$

The oracle bound uses half the trace of $\boldsymbol{C}$ over the discarded subspace. For comparison, let $\boldsymbol{P}_r^{\boldsymbol{C}} := \boldsymbol{V}_r^{\boldsymbol{C}}\boldsymbol{V}_r^{\boldsymbol{C}\top}$ project onto the top-r eigenspace of $\boldsymbol{C}$. Subtracting and adding the trace at this projector separates the rank and suboptimality contributions:

$$C_1 \;=\; \underbrace{\tfrac{1}{2}\textstyle\sum_{i>r}\lambda_i(\boldsymbol{C})}_{\textit{rank}} \;+\; \underbrace{\tfrac{1}{2}\big[\mathrm{Tr}(\boldsymbol{P}_r^{\boldsymbol{C}}\boldsymbol{C}) - \mathrm{Tr}(\widehat{\boldsymbol{P}}_r\boldsymbol{C})\big]}_{\Delta_{\mathrm{subopt}}\,\ge\,0}, \tag{3.5}$$

where the rank term is fixed by $\boldsymbol{C}$ and r. The remaining term Δ_{subopt} is of order $\|\sin\Theta(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}, \boldsymbol{V}_r^{\boldsymbol{C}})\|_F^2$. It is zero whenever $\mathrm{span}(\widehat{\boldsymbol{V}}_r^{\boldsymbol{C}}) = \mathrm{span}(\boldsymbol{V}_r^{\boldsymbol{C}})$, and only in this case if $\lambda_r(\boldsymbol{C}) > \lambda_{r+1}(\boldsymbol{C})$.

To prove Proposition 3.1, apply the triangle inequality in $L^2(\pi_{\boldsymbol{Z}};\mathbb{R}^d)$ to the Gaussian-reference bound (2.5) of Part I (Appendix A). This bound is [36, Cor. 2.10] specialized to a standard Gaussian reference. In the score-ratio-matching stability result of [5, Thm. 3.3], the divergence at the top-r eigenspace of $\boldsymbol{C}'$ is bounded by $\|\mathcal{S}-\mathcal{S}'\|^2_{L^2(\pi_{\boldsymbol{Z}})} + \sum_{i>r}\lambda_i(\boldsymbol{C}')$. This follows from (3.3) at that eigenprojector after applying $(a+b)^2 \le 2a^2+2b^2$. Our bound retains the square root and applies to the estimated projector $\widehat{\boldsymbol{P}}_r$.

Projecting $\mathcal{S}$ onto the gradient space $\mathcal{F}_\Lambda$ in $L^2(\pi_{\boldsymbol{Z}})$ defines $\mathcal{S}_\Lambda$, its second-moment matrix $\boldsymbol{C}_\Lambda$, and the top-r projector $\boldsymbol{P}_r^{\Lambda\star}$. Because $\widehat{\boldsymbol{C}}$ is computed from the penalized